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Studierichting: Informatica  
Oefeningenreeks 6B nr. 5.8

$$x_1 \text{ als } s_2 = -4 \text{ en } s_6 = -84$$

$$s_2 = x_1 \frac{1 - q^2}{1 - q} = -4$$

$$= x_1(1 + q) = -4$$

$$s_6 = x_1 \frac{1 - q^6}{1 - q} = -84$$

$$= x_1(1 + q + q^2 + q^3 + q^4 + q^5) = -84$$

$$\Rightarrow \frac{x_1(1 + q + q^2 + q^3 + q^4 + q^5)}{x_1(1 + q)} = \frac{-84}{-4}$$

$$\Rightarrow \frac{x_1(1 + q)(1 + q^2 + q^4)}{x_1(1 + q)} = 21$$

$$\Rightarrow 1 + q^2 + q^4 = 21$$

$$y = q^2$$

$$y^2 + y + 1 = 21$$

$$y^2 + y - 20 = 0$$

$$\Delta = 1^2 - 4 \cdot 1 \cdot (-20) = 81$$

$$y_{1,2} = \frac{-1 \pm \sqrt{81}}{2}$$

$$\begin{cases} y_1 = 4 \\ y_2 = -5 \end{cases}$$

$$\Rightarrow q^2 = 4 \Rightarrow q = \pm 2$$

$$q^2 = -5 \Rightarrow q \notin \mathbb{R}$$

$$\Rightarrow \text{Als } q = 2 :$$

$$x_1(1 + 2) = -4$$

$$3x_1 = -4$$

$$x_1 = \frac{-4}{3}$$

$$\text{Als } q = -2 :$$

$$x_1(1 + (-2)) = -4$$

$$-x_1 = -4$$

$$x_1 = 4$$

## References